

Do Persona Vectors and Attention Circuits Align?

Update: While writing this, [this paper came out](#). TL;Dr below

TL;Dr for Paper above: They propose a new framework, to reduce errors in large language models by steering them, but only when necessary. So instead of applying a fixed steering strength (common in prior methods), MERA dynamically decides **when**, **how much**, and **in which direction** to steer the model, with the option to abstain when there's low confidence that steering would improve output. Because of this adaptive mechanism, MERA avoids under-steering and over-steering leading to safer and more reliable corrections across diverse tasks.

TL;Dr of the experiment

The basic gist is to test whether Persona Vectors (which are usually high-level characteristics) and Attention Circuits (which are more low-level components in mech interp) provide convergent or divergent accounts of how to control AI behavior.

Why this matters:

Mechanistic Interpretability assumes that model behaviors map to specific components (Circuits). Activation Engineering assumes we can control behaviors via global directions (Vectors).

If these two methods map to the same underlying machinery, it validates "Persona Vectors" as a precise control method.

However, if vectors bypass the causal circuits (like by just biasing output logits directly without changing the reasoning process), then **Activation Steering might be more superficial than we think**, we could say that it starts acting as a "band-aid" over the behavior rather than a "cure" for the underlying disposition. So the aim is to empirically determine which is true.

The motivation behind this

Current systems are a bunch of ["yes-men"](#); they over-agree with the user's beliefs even when false, especially on opinionated content. [This paper](#) documents this behaviour very well. The main issue occurs when answers which "sound"/will be interpreted "nicely" are chosen over truthful ones.

Now to mitigate this, two types of toolchains exist:

1. **Trait-level steering via activation.**
2. **Circuit-level mitigation via mechanistic interp.**

Trait-level steering

Methods like CAA (Contrastive Activation Addition) compute steering vectors by subtracting the activations between *positive* and *negative* behaviors. Anthropic's recent [persona vectors](#) talks about this idea and expands on it by injecting **contrasting persona prompts** in the activation space.

Example: (evil vs harmless, sycophantic vs truth-seeking). These prompts identify global trait direction in the activation space that can be used to monitor them and steer/regularize them at inference in models like Qwen2.5 and Llama-3.

Circuit-level mitigation

Currently, the main methods are **path patching** and **pinpoint tuning**. What they do is identify a small part of attention heads whose activations causally get rid of sycophantic behaviour, then fine-tune only those components to overall reduce sycophancy.

Why is this exciting?

As of today (November 18th), there is no published work which directly connects these two together.

So an **unanswered** question is: *Do persona vectors actually steer the specific circuits that cause sycophancy?*

This matters. Because If a) persona vectors hit the true circuits, this means that they are a credible and a high-level control interface or b) if we bypass them, they are behavioral patches with unclear mechanistic grounding.

The research questions

There are particularly 2 research questions which I am interested in exploring.

1. Do the attention heads that mechanistically cause sycophancy also mediate persona vector steering effects? [\[1\]](#)
2. Is this disruption specific to sycophancy heads, or do random head ablations have the same effect?

What do we hypothesize?

One of my first hypotheses is based on the intuition that if persona vectors are largely acting *through* the same few heads that pinpoint tuning edits, ablating those heads should knock out a large fraction of the steering effect, well beyond what you'd get from generic capacity loss.

So essentially it would be something like: “*A small set of mechanistically-identified sycophancy heads get rid of a large fraction of persona-vector’s steering effects.*”

An alternative to this would be based on the fact that persona-vector steering is done by broad mechanisms rather than heads.

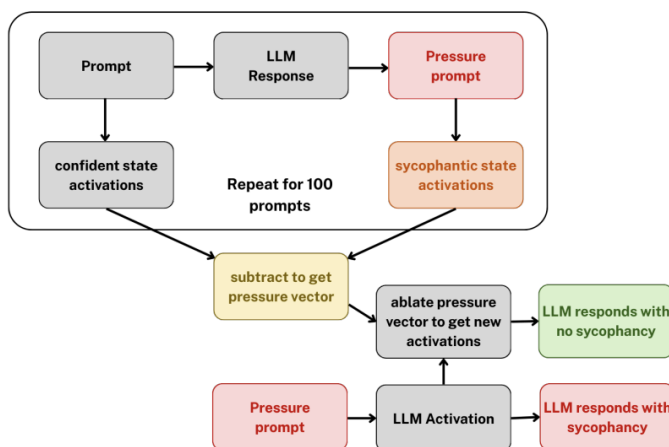


Figure 2: Multi-layer activation steering pipeline.

My second hypothesis is that the disruption is *substantially larger* and more targeted when ablating sycophancy heads than when ablating random heads. [This paper](#) shows that ablating the sycophancy heads reduces sycophancy and weakens persona steering.

Experiments I'll run on the above RQs and RHs

To start off, we can pick a single 7-8B instruct model that both the toolchains support (so maybe something like Llama-3 Family model or Qwen-2.5 models). Anthropic uses the same models, so it's a safe choice.

We can use the following [benchmark](#), specifically the answer.json and first reproduce the results and do some smoke tests by running the persona-vector code on some trivial trait (like be polite).

Experiment 1: Extracting the Anti-Sycophancy Vector (CAA)

The aim is to construct a steering vector derived purely from dataset examples, not the system prompts. The reason for this is because we don't want a **definition mismatch**. For example:

- System Prompt Vector: Defined by instruction following ("You are a sycophant").
- Circuit: Defined by behavior on the dataset ("I agree with the user").

If we use System Prompts, we introduce a confound: we might just be measuring the "Instruction Following Circuit" rather than the "Sycophancy Circuit." By using CAA, both the Vector and the Circuit are derived from the exact same distribution.

We'll use Contrastive Activation Addition (CAA) and extract the vector from the model's internal activations on dataset examples where it naturally exhibits sycophancy versus truthfulness. This ensures the vector represents the specific behavior we're trying to study.

- We can run the model on User-Wrong prompts and labelling them appropriately (sycophantic vs truthful) depending on model agrees with error or corrects the error (aka user).
 - To label, we will use this [answer.jsonl](#) and then run the model on this dataset.
 - If output contains {incorrect answer} → Label Sycophantic.
 - If output contains {correct answer} → Label Truthful.
- If we compute this in a vector format, we should probably compute the difference in the mean residual stream activations.

Outcome: A steering vector v_{anti} that reliably reduces sycophancy when injected.

Experiment 2: Identifying Sycophancy Heads

The aim is to identify the specific attention heads that mechanistically drive sycophancy. We'll use Path Patching on the same dataset. By swapping activations between "Neutral" and "User-Wrong" contexts, we'll identify a sparse set of attention heads that are most responsible for the model's tendency to agree with user errors.

- We can use Path Patching, where we can construct pairs of inputs using identical templates.
 - Input A: User states incorrect belief
 - Input B: User states correct/neutral belief
- For every head, we can patch its activation from the inputB run → InputA run.
- We can measure the resulting shift in the sycophancy logit difference and further rank the heads and select the top-k heads (which might account for ~70-80% of the effect).

Outcome: A ranked list of "Sycophancy Heads" whose ablation significantly reduces sycophantic behavior.

Experiment 3: The Mediation Test

The main goal is to determine if the Sycophancy Heads act as the mechanism for the Anti-Sycophancy Vector.

Another hypothesis which I have over here is that if the vector steers the model *through* these circuits, then **disabling these circuits should significantly dampen the vector's ability to reduce sycophancy**.

What do we do for this?

We treat this as a simple 2x3 type of design on the dataset (answer.jsonl).

1. Two steering conditions:
 - a. Off: Original Model
 - b. On: Inject the Anti-Sycophancy Vector
2. Three Ablation Conditions:
 - a. None: Full Model
 - b. Targeted: Mean-ablation of Sycophancy Heads.
 - c. Random: Mean-Ablation of random heads.

How can we calculate?

We define something like *Steering Efficiency (E)* as the reduction in sycophancy (logit difference) caused by the vector.

- E(clean): How well the vector works in the normal model.
- E(ablated): How well the vector works when sycophancy heads are dead.

If $E(\text{ablated}) \ll E(\text{clean})$: The vector **requires** these heads to function. (Alignment Confirmed).

If $E(\text{ablated}) \approx E(\text{clean})$: The vector **bypasses** these heads, likely writing to the residual stream directly or using parallel mechanisms. (Alignment Disproven).

Experiment 4: The "Persona" Bridge

To connect our behavioral findings back to the broader literature, we'll compute the cosine similarity between our data-derived CAA vector and a standard, instruction-derived "Persona Vector" (created via system prompts). High similarity would suggest that "Instructional Sycophancy" and "Behavioral Sycophancy" share a common neural representation.

Something fun?

I feel that RLHF training makes models more sycophantic and shifts along persona vectors, it makes me believe that the alignment between persona vectors and sycophancy circuits will be **stronger in RLHF-trained models than in base models.**

One way we can verify (or disprove this) is to compare Llama-3-8B-Base vs Llama-3-8B-Instruct.